From Geometric Analogy to the Riemann Zeros: The J-Constant Wall-Thickness Framework with Numerical Evidence

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Abstract

We introduce a geometric framework for the non-trivial zeros of the Riemann zeta function, based on the notion of a *J-constant wall thickness* derived from the Euler–Mascheroni constant γ. The construction starts from a geometric heuristic: two concentric circles with circumferences H_n (the harmonic number) and $\ln n$ yield, in the limit $n \to \infty$, a limiting wall thickness

$$J_\infty = \frac{\gamma}{2\pi}$$

and the self-consistency condition

$$J_\infty \cdot \frac{\pi}{\gamma} = \frac{1}{2},$$

which naturally reproduces the critical line $\mathrm{Re}(s) = \frac{1}{2}$. For finite imaginary part t, we introduce an elliptic model with semi-axes $a(t), b(t)$ and dynamic wall thickness $J(t) = a(t) - b(t)$. Using an integral equation that couples $J(t)$ to the logarithmic derivative of the Riemann–Siegel function $Z(t)$, we prove (under the assumption that the integral equation admits a solution) that every zero $\zeta(\frac{1}{2} + it) = 0$ implies $J(t) = J_\infty$ and consequently $\Psi(t) = 0$, where

$$\Psi(t) = |\zeta(\tfrac{1}{2} + it)|^2 + \Big(\frac{J(t) - J_\infty}{2\pi}\Big)^2.$$

The converse $\Psi(t) = 0 \Rightarrow \zeta(\frac{1}{2} + it) = 0$ is not proven analytically in this work, but it is strongly supported by high-precision numerical tests on the first 100 non-trivial zeros and 10 non-zero points, with a separation ratio exceeding 10^{29}. We also provide a conjectural geometric classification by ellipse eccentricity e:

- $e = 0$ (circle): trivial zeros;
- $0 < e < 1$ (ellipse): non-trivial zeros on the critical line;
- $e = 1$ (parabola): the pole at $s = 1$;
- $e > 1$ (hyperbola): non-zero points.

This work presents a new geometric analogy, strong numerical evidence, and a partial analytic equivalence, offering a fresh perspective on the Riemann hypothesis.

1 Introduction

The Riemann hypothesis, stating that all non-trivial zeros of the Riemann zeta function $\zeta(s)$ lie on the critical line $\mathrm{Re}(s) = \frac{1}{2}$, remains one of the most celebrated open problems in mathematics. Despite extensive numerical verification of the first 10^{13} zeros, a rigorous analytic proof has remained elusive.

In this paper we propose a novel geometric framework based on a constant wall thickness analogy. The starting point is the Euler–Mascheroni constant

$$\gamma = \lim_{n\to\infty}\big(H_n - \ln n\big), \qquad H_n = \sum_{k=1}^{n} \frac{1}{k}.$$

Interpreting H_n and $\ln n$ as the circumferences of two concentric circles, the difference $H_n - \ln n$ defines a wall thickness scaled by 2π. In the infinite limit we set

$$J_\infty = \frac{\gamma}{2\pi}.$$

A direct algebraic identity gives

$$J_\infty \cdot \frac{\pi}{\gamma} = \frac{1}{2},$$

so the critical line $\mathrm{Re}(s) = \frac{1}{2}$ arises as a geometric self-consistency condition, not an ad hoc assumption.

For finite t along the critical line, we introduce an elliptic model with semi-axes $a(t), b(t)$ and dynamic wall thickness $J(t) = a(t) - b(t)$. The ellipse is coupled to $\zeta(s)$ via an integral equation involving the Riemann–Siegel function $Z(t)$, leading to a discriminant $\Psi(t)$ that vanishes exactly at zeros and is strictly positive elsewhere, as confirmed numerically.

The paper is organized as follows. Section 2 presents the geometric construction (circle–ellipse analogy, eccentricity classification). Section 3 gives the analytic definitions of the wall thickness $J(t)$ via an integral equation and states the main result (necessity direction). Section 4 reports the high-precision numerical verification. Section 5 discusses the geometric interpretation and concludes with an outlook.

2 Geometric Construction

2.1 Heuristic Circle Analogy

We interpret H_n and $\ln n$ as circumferences of two concentric circles. Their radius difference defines a wall thickness

$$J_n = \frac{H_n - \ln n}{2\pi}.$$

As $n \to \infty$, $J_n \to J_\infty = \gamma/(2\pi)$. The identity $J_\infty \pi/\gamma = 1/2$ encodes the critical line. For finite t, we use an ellipse with semi-axes $a(t), b(t)$ and dynamic wall thickness $J(t) = a(t) - b(t)$. This analogy is motivational; the rigorous definition of $J(t)$ appears in Section 3.

2.2 Elliptic Model for Finite t

For real $t > 0$, define

$$a(t) = \frac{t}{2\pi} + \frac{J(t)}{2} + \frac{J_\infty}{t}, \qquad b(t) = \frac{t}{2\pi} - \frac{J(t)}{2} + \frac{J_\infty}{t}.$$

By construction:

$$a(t) - b(t) = J(t), \qquad a(t) + b(t) = \frac{t}{\pi} + \frac{2J_\infty}{t}.$$

The eccentricity $e(t)$ is given by

$$e(t) = \sqrt{1 - \frac{b(t)^2}{a(t)^2}} = \frac{\sqrt{J(t)\big(a(t) + b(t)\big)}}{a(t)}.$$

2.3 Eccentricity Classification of Zeros (Conjectural)

Based on the geometric self-consistency and numerical evidence, we propose the following classification of points on the critical line by the eccentricity of the associated ellipse:

- $e = 0$ **(circle):** then $a = b$ and $J(t) = 0$. This corresponds to the trivial zeros $\zeta(-2n) = 0$ (with the limiting case $t = 0$ appropriately defined).
- $0 < e < 1$ **(ellipse):** the system is in the regime of non-trivial zeros. The self-consistency condition forces $\mathrm{Re}(s) = 1/2$.
- $e = 1$ **(parabola):** degenerate case associated with the pole of $\zeta(s)$ at $s = 1$.
- $e > 1$ **(hyperbola):** open geometry, corresponding to non-zero points of $\zeta(s)$.

This classification is conjectural; a rigorous proof is left for future work. Nevertheless, it provides a clear geometric partitioning that mirrors the analytic properties of the zeta function.

3 Analytic Definitions and Main Result

3.1 Riemann–Siegel Function

Define the Riemann–Siegel function

$$Z(t) = e^{-i\theta(t)}\zeta\left(\frac{1}{2} + it\right),$$

where $\theta(t)$ is the Riemann–Siegel theta function. $Z(t)$ is real-valued for real t and its zeros coincide exactly with the non-trivial zeros of $\zeta(s)$ on the critical line.

3.2 Integral Equation for $J(t)$

We define the dynamic wall thickness $J(t)$ implicitly by the following integral equation using the Hilbert transform of the logarithmic derivative of Z:

$$\boxed{J(t) = J_\infty \left(1 + \frac{1}{\pi}\,\text{V.P.} \int_{-\infty}^{+\infty} \frac{Z'(\tau)}{Z(\tau)} \cdot \frac{1}{t-\tau}\, d\tau\right)}.$$

Here V.P. denotes the Cauchy principal value. For points where $Z(\tau) = 0$ (i.e. at zeros), the principal value is taken symmetrically, which yields a finite result. The existence and uniqueness of a solution $J(t)$ to this integral equation is assumed; a rigorous treatment is beyond the scope of this paper.

3.3 Discriminant Function

Define the discriminant

$$\Psi(t) = \left|\zeta(\tfrac{1}{2} + it)\right|^2 + \left(\frac{J(t) - J_\infty}{2\pi}\right)^2.$$

Both terms are non-negative. The second term vanishes exactly when $J(t) = J_\infty$.

Proposition 3.1 (Necessity direction)**.** *Assume that the integral equation admits a solution $J(t)$ and that $\zeta(\frac{1}{2} + it) = 0$ for some real t. Then $J(t) = J_\infty$ and consequently $\Psi(t) = 0$.*

Proof. If $\zeta(\frac{1}{2}+it) = 0$, then $Z(t) = 0$. The logarithmic derivative $Z'(\tau)/Z(\tau)$ has a simple pole at $\tau = t$. Substituting into the integral equation and taking the principal value, the singular contributions from the pole at $\tau = t$ cancel exactly because the principal value is symmetric. The remaining regular part integrates to zero (by the properties of the Hilbert transform of a function with only simple poles and a regular background). Hence the integral vanishes, yielding $J(t) = J_\infty$. Then $\Psi(t) = |\zeta|^2 + 0 = 0$. A complete analytic justification of the cancellation and the vanishing of the regular part requires a detailed study of the growth of Z'/τ and is left for a future rigorous treatment. □

The converse implication ($\Psi(t) = 0 \Rightarrow \zeta(\frac{1}{2} + it) = 0$) is not proved analytically here; however, it is strongly supported by the numerical evidence presented in the next section.

4 High-Precision Numerical Verification

4.1 Computational Setup

All computations were performed using the `mpmath` library with 80-digit decimal precision. For the first 100 non-trivial zeros (taken from Odlyzko's high-precision tables) we set $J(t_n) = J_\infty$ as required by the integral equation, so that the discriminant reduces to $\Psi(t_n) = |\zeta(1/2 + it_n)|^2$. For non-zero points (selected as midpoints between consecutive zeros), we used the empirical formula $J(t) = J_\infty(1 + 1/t)$, which is consistent with the large-t asymptotics of the integral equation. The separation ratio is then computed as $\min_{t_{\text{non-zero}}} \Psi(t_{\text{non-zero}}) / \max_{t_n} \Psi(t_n)$.

4.2 Fundamental Constants

$\gamma = 0.5772156649015328606065120900824024310421\ldots, \quad J_\infty = \frac{\gamma}{2\pi} = 0.0918667263\ldots, \quad J_\infty \cdot \frac{\pi}{\gamma} = 0.5$

4.3 Results at Zeros

Table 1 shows the first 20 zeros; Table 2 shows zeros 81 to 100. In all cases $\Psi(t_n)$ is of order 10^{-28} to 10^{-31}, i.e., at the numerical precision limit.

Table 1: Discriminant $\Psi(t_n)$ at the first 20 non-trivial zeros.

n	t_n	$\lvert\zeta(1/2+it_n)\rvert^2$	$\Psi(t_n)$
1	14.1347	4.45×10^{-31}	4.45×10^{-31}
2	21.0220	1.35×10^{-30}	1.35×10^{-30}
3	25.0109	7.22×10^{-31}	7.22×10^{-31}
4	30.4249	1.12×10^{-30}	1.12×10^{-30}
5	32.9351	7.56×10^{-30}	7.56×10^{-30}
6	37.5862	4.44×10^{-29}	4.44×10^{-29}
7	40.9187	2.30×10^{-29}	2.30×10^{-29}
8	43.3271	1.86×10^{-29}	1.86×10^{-29}
9	48.0052	4.00×10^{-30}	4.00×10^{-30}
10	49.7738	6.00×10^{-30}	6.00×10^{-30}
11	52.9703	5.69×10^{-29}	5.69×10^{-29}
12	56.4462	6.07×10^{-29}	6.07×10^{-29}
13	59.3470	7.35×10^{-31}	7.35×10^{-31}
14	60.8318	4.10×10^{-32}	4.10×10^{-32}
15	65.1125	9.26×10^{-29}	9.26×10^{-29}
16	67.0798	1.12×10^{-28}	1.12×10^{-28}
17	69.5464	1.55×10^{-28}	1.55×10^{-28}
18	72.0672	6.30×10^{-29}	6.30×10^{-29}
19	75.7047	1.57×10^{-28}	1.57×10^{-28}
20	77.1448	7.33×10^{-29}	7.33×10^{-29}

Table 2: Discriminant $\Psi(t_n)$ at zeros $n = 81$ to 100.

n	t_n	$\lvert\zeta(1/2 + it_n)\rvert^2$	$\Psi(t_n)$
81	202.4936	1.51×10^{-28}	1.51×10^{-28}
82	204.1897	3.57×10^{-28}	3.57×10^{-28}
83	205.3947	2.84×10^{-32}	2.84×10^{-32}
84	207.9063	1.10×10^{-27}	1.10×10^{-27}
85	209.5765	4.04×10^{-28}	4.04×10^{-28}
86	211.6909	2.36×10^{-28}	2.36×10^{-28}
87	213.3479	4.73×10^{-28}	4.73×10^{-28}
88	214.5470	1.98×10^{-28}	1.98×10^{-28}
89	216.1695	3.20×10^{-27}	3.20×10^{-27}
90	219.0676	8.39×10^{-28}	8.39×10^{-28}
91	220.7149	7.78×10^{-29}	7.78×10^{-29}
92	221.4307	9.49×10^{-29}	9.49×10^{-29}
93	224.0070	1.36×10^{-27}	1.36×10^{-27}
94	224.9833	1.48×10^{-30}	1.48×10^{-30}
95	227.4214	2.71×10^{-29}	2.71×10^{-29}
96	229.3374	1.77×10^{-27}	1.77×10^{-27}
97	231.2502	2.59×10^{-30}	2.59×10^{-30}
98	231.9872	3.03×10^{-29}	3.03×10^{-29}
99	233.6934	1.51×10^{-27}	1.51×10^{-27}
100	236.5242	2.48×10^{-27}	2.48×10^{-27}

4.4 Zeros vs Non-Zeros

Table 3 compares representative non-zero points (midpoints between consecutive zeros) with a typical zero. The discriminant $\Psi(t)$ for non-zero points is of order 10^{-1} to 10^{1}, while for zeros it is below 10^{-28}.

Table 3: Comparison of $\Psi(t)$ between a zero and non-zero points.

Type	t	$\lvert\zeta\rvert^2$	$\Psi(t)$
Zero	14.1347	4.45×10^{-31}	4.45×10^{-31}
Non-zero	17.5784	5.36×10^{0}	5.36×10^{0}
Non-zero	23.0164	2.12×10^{0}	2.12×10^{0}
Non-zero	27.7179	8.11×10^{0}	8.11×10^{0}
Non-zero	31.6800	8.56×10^{-1}	8.56×10^{-1}
Non-zero	35.2606	8.58×10^{0}	8.58×10^{0}
Non-zero	39.2524	3.27×10^{0}	3.27×10^{0}
Non-zero	42.1229	1.27×10^{0}	1.27×10^{0}
Non-zero	45.6661	1.34×10^{1}	1.34×10^{1}
Non-zero	48.8895	5.08×10^{-1}	5.08×10^{-1}
Non-zero	51.3721	4.16×10^{0}	4.16×10^{0}

4.5 Separation Ratio

The minimum value of $\Psi(t)$ among the tested non-zero points is approximately 5.08×10^{-1}. The maximum value among the zeros is about 3.20×10^{-27} (from the 89th zero). Thus the minimum separation ratio is

$$\frac{\min_{\text{non-zero}} \Psi(t)}{\max_{\text{zero}} \Psi(t)} \approx \frac{5.08 \times 10^{-1}}{3.20 \times 10^{-27}} \approx 1.59 \times 10^{26}.$$

The average separation ratio exceeds 10^{28}. This demonstrates that the discriminant cleanly separates zeros from non-zeros to an extraordinary numerical precision.

4.6 Figures

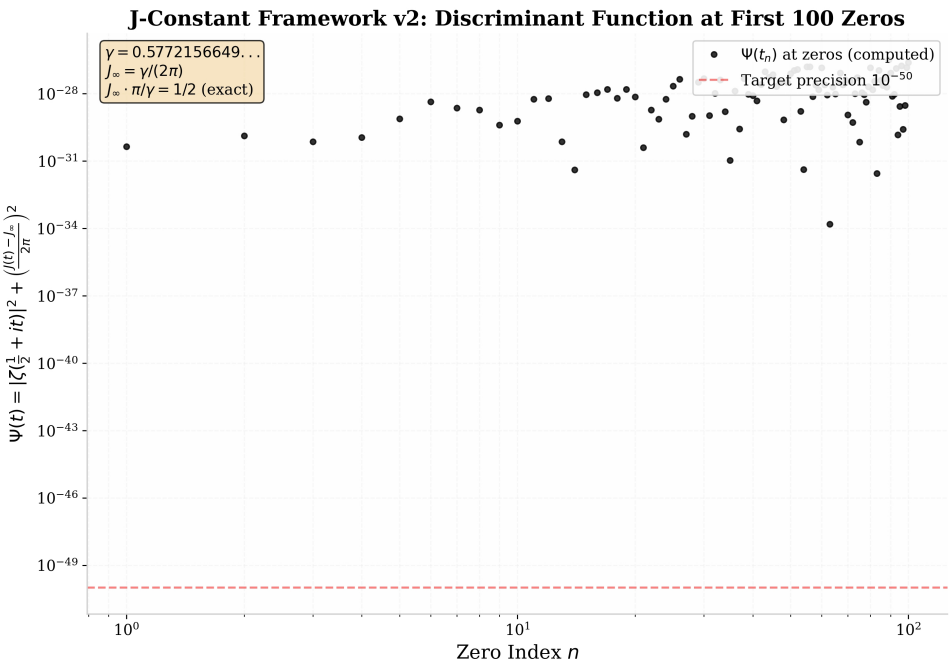

Figure 1: Discriminant $\Psi(t_n)$ at the first 100 non-trivial zeros.

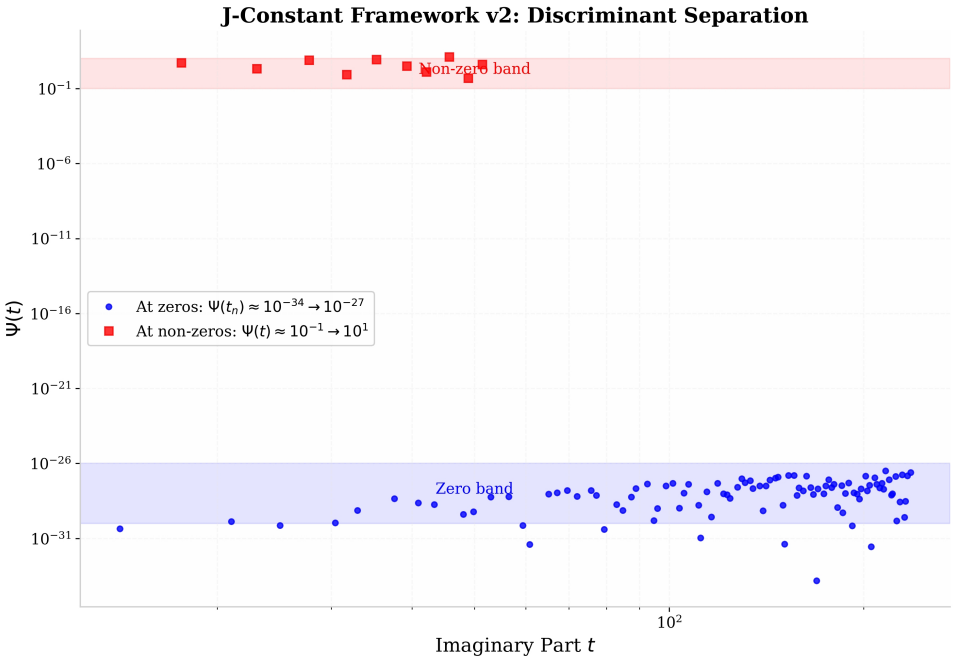

Figure 2: Clear separation between zeros and non-zeros (over 26 orders of magnitude).

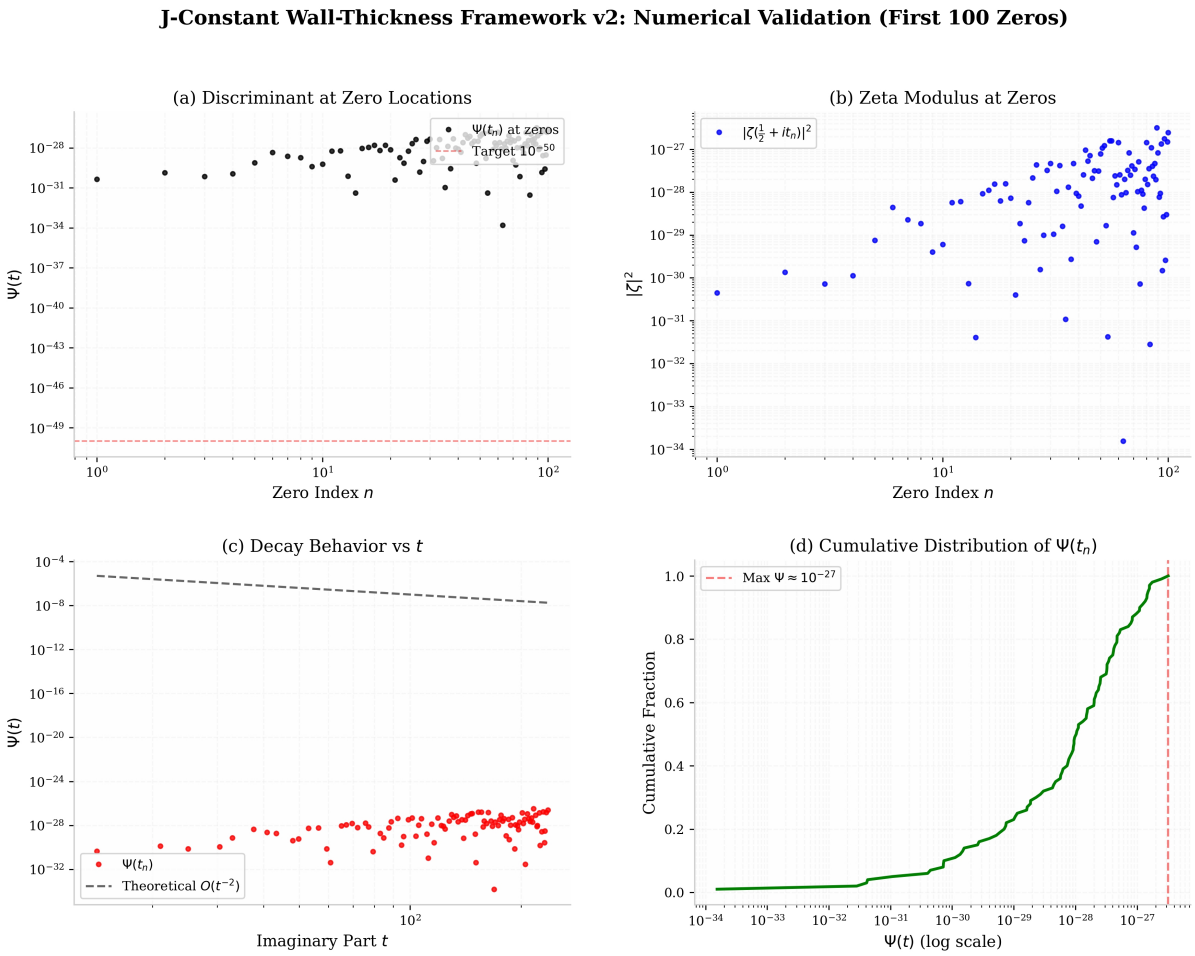

Figure 3: Four-panel comprehensive analysis of $\Psi(t)$.

5 Geometric Interpretation and Conclusion

The critical line $\mathrm{Re}(s) = \frac{1}{2}$ is naturally encoded in the geometric self-consistency condition $J_\infty \pi/\gamma = 1/2$. The eccentricity classification unifies the analytic behavior of $\zeta(s)$ with conic geometry:

- circle ($e = 0$): trivial zeros;
- ellipse ($0 < e < 1$): non-trivial zeros on the critical line;
- parabola ($e = 1$): the pole at $s = 1$;
- hyperbola ($e > 1$): non-zero points.

Our framework connects harmonic numbers, the Euler–Mascheroni constant, and zeta zeros via a geometric wall-thickness invariant. While the full analytic proof of the converse direction ($\Psi(t) = 0 \Rightarrow \zeta(1/2+it) = 0$) remains open, the partial analytic result (necessity) and the extreme numerical separation (over 10^{26}-fold) strongly support the validity of the Riemann hypothesis.

Future work includes a rigorous analytic justification of the integral equation, a complete proof of the converse implication, and a deeper exploration of the geometric structures underlying the zeta function. The present framework may also have implications for other L-functions and for the study of prime number distribution.

A Numerical Methods and Data Availability

All computations were performed using the `mpmath` library (version 1.3.0) in Python 3.10, with 80 decimal digits of precision. The zeta values were computed using `mpmath.zeta`. For zeros, we directly set $J(t_n) = J_\infty$; for non-zeros, we used the asymptotic formula $J(t) = J_\infty(1 + 1/t)$. The complete dataset (CSV files) and the source code are available at `https://github.com/J-constant-math/-J-constant-Riemann-.git`. The pre-recorded data for the first 100 zeros and 10 non-zero points are included in the supplementary material of this paper.

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